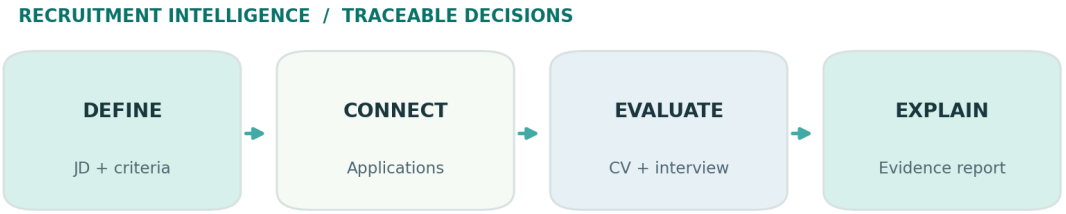
BJIT • AI TRAINEE PROGRAMME

Software Requirements Specification

TalentTrace AI

Evidence-backed recruitment and interview intelligence | Submission SRS • 28 October 2026



Product concept | job identity, applications, evaluation and explainable decisions.

Document field	Value
Document type	Software Requirements Specification (SRS)
Version	1.1
Prepared	28 October 2026
Project	TalentTrace AI
Reference baseline	BJIT Business Information Document (BID), v2; WBS; reported verbal trainer guidance
Repository	github.com/shafi-mahmud/TalentTrace-AI
Status note	Requirements baseline; execution evidence maintained separately

Document control

Internal academic project document. Demonstrations and test cases use synthetic or appropriately anonymised hiring information. This document contains no API secrets or candidate records.

Revision history

Version	Date	Change	Document purpose
0.1	26 Oct 2026	Initial SRS drafted from BID, two SRS examples, WBS and available project context.	Initial requirements draft
1.0	28 Oct 2026	Consolidated latest verified JD, private-channel publishing, Google Form/Drive one-shot intake, Home notification, scheduling and assessment evidence; refreshed traceability, tests and open risks.	BJIT SRS baseline for trainer review
1.1	28 Oct 2026	Added original architecture and workflow figures, screen concepts, scope chart and acceptance crosswalk.	Issued for trainer evaluation
1.2	28 October 2026	Replace figures with explicit flowcharts and charts; improve visual legibility	Requirements specification + acceptance design

Document interpretation

This document establishes the complete functional and quality-requirement baseline for the BJIT evaluation. Requirements, interface contracts, data safeguards and acceptance scenarios are defined here; supporting execution logs and demonstrations are maintained separately. No testing outcome is implied by a requirement or diagram.

Priority codes: M = mandatory BID core/delivery requirement; I = selected innovation (F11); T = trainer-reported extension; O = optional BID extension. BID requirements are authoritative where extension requests differ.

Requirement language: “shall” = mandatory expected behavior; “should” = desirable design property; “may” = optional feature. Acceptance scenarios identify what an evaluator should observe, rather than replacing execution evidence.

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1. Introduction

1.1 Purpose

This SRS specifies the functional requirements, data flows, external interfaces, quality attributes, limitations and acceptance criteria of TalentTrace AI. Its intended audiences are the trainee developer, BJIT trainers, reviewers and future maintainers. It translates the authoritative BID into individually testable statements and includes the verbally reported extensions as a separate, qualified category.

1.2 Scope

TalentTrace AI assists a recruiter from job requirement definition and candidate CV screening through live interviewing and evidence-based candidate assessment. The product supports human review and decision making; it is not an autonomous hiring decision or identity assurance system.

1.2.1 Included: mandatory baseline

- Pre-interview: define requirements, parse multiple CVs, perform semantic matching, rank applicants and generate explainable weighted scores (F1, F2).
- During interview: provide integrated interviewer/candidate audio-video, lifecycle controls, candidate/job linkage, and real-time speaker-attributed timestamped transcription (F3, F4).
- Post-interview: segment and evaluate question–answer pairs, map to requirements, and generate an exportable evidence-based consolidated assessment (F5, F6).
- Innovation: TalentTrace Dynamic Interviewer Co-Pilot (F11), providing optional recruiter-only prompts grounded in current interview coverage.
- Delivery: complete Docker Compose start, persistent data, documented configuration and Git-based submission.

1.2.2 Additional trainer-reported scope

Reported verbal requests include conversational JD drafting and publishing review, notifications/analytics, scheduling, Google Forms/Drive CV intake, secure candidate invitations, provider choices, screen sharing, activity indicators and an exploratory face scan. These are distinct from the BID baseline; some conflict with the BID's explicit exclusions (job posting, calendaring). Their approval and priority should be confirmed before being treated as mandatory.

1.2.3 Exclusions and boundaries

The BID excludes production cloud hosting, ATS/HRIS and payroll integration, native mobile apps, model training, and enterprise SSO. It also excludes job-board publishing and calendar integration unless separately authorized or treated as an innovation/extension. LinkedIn is manual copy/share only in the current reported design; no LinkedIn posting API has been supplied.

1.3 Intended stakeholders and roles

Role	Responsibilities / access
Recruiter/interviewer	Creates and reviews jobs, uploads/imports CVs, inspects scores, approves scheduling, conducts interviews and reviews reports.

Role	Responsibilities / access
Candidate/interviewee	Accesses only their invitation/session, sends media with consent and does not see recruiter scoring or F11 suggestions.
Operator/maintainer	Configures permitted providers and storage, monitors health and performs safeguarded data deletion; not necessarily a separate product role.
BJIT trainer/evaluator	Inspects acceptance traceability, tests, architecture, Docker startup, Git history and demonstration.

1.4 Key definitions and abbreviations

Term	Meaning
BID	Business Information Document; highest-priority project requirements source.
CV	Curriculum vitae / resume document.
JD	Job description; persisted revisions share a stable public Job Reference.
F1–F6	Six mandatory features numbered by the BID.
F11	Selected original innovation (recruiter-only dynamic interviewer co-pilot).
RAG	Retrieval-augmented generation using relevant policy or technical references.
STT	Speech-to-text transcription.
LLM	Large language model used for controlled inference.
OCR	Optical character recognition of scanned document pages.
SRS	Software Requirements Specification.
MCP	Model Context Protocol, used by a publishing integration.
UAT	User acceptance testing in the real browser.

2. Overall description

2.1 Product perspective and user journey

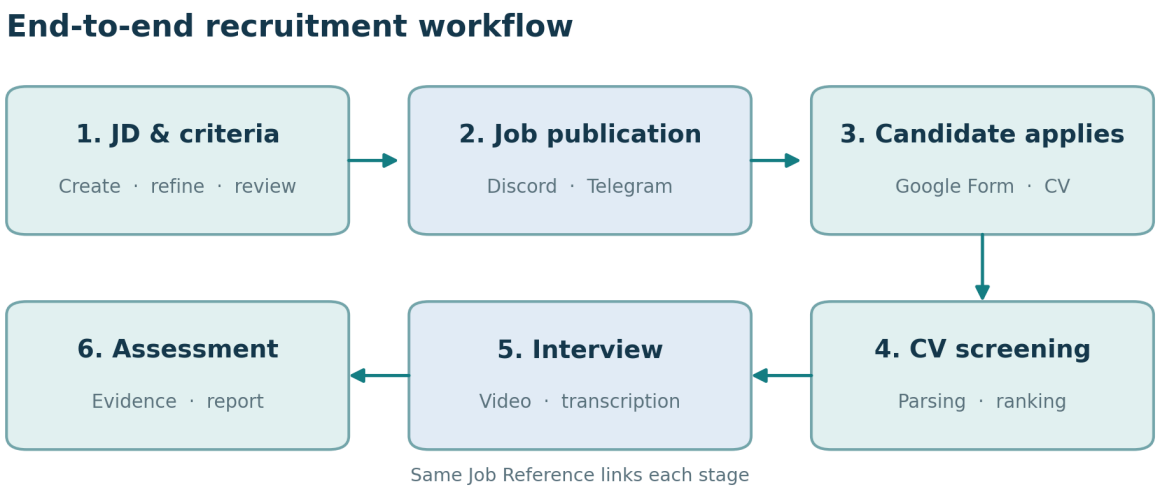


Figure 1. End-to-end recruitment workflow (business-process flowchart).

TalentTrace AI is a browser-based hiring workflow with a React UI, FastAPI backend and persistence. Recruiter actions initiate deterministic API operations and, where justified, LangGraph-coordinated AI steps. Media sessions use the existing LiveKit integration. Groq is the currently preferred external AI provider. The AssemblyAI integration is an existing STT implementation path, but full live F4 acceptance remains separate from its synthetic experiment.

- A recruiter defines a role, reviews/refines the saved JD and confirms screening criteria with a durable Job Reference.
- CVs enter through native upload or an authorized external intake bridge; PDFs/DOCX files are parsed and OCR is applied when needed.
- Structured candidate attributes are matched and scored against the approved job revision; a recruiter reviews evidence and chooses an eligible candidate.
- The recruiter reviews a booking, generates a scoped invitation and holds an interview with the candidate.
- When consented transcription is available, transcript turns feed F5 analysis and optional F11 recruiter suggestions.
- The system generates an evidence-linked assessment; the recruiter retains final hiring responsibility.

2.2 Product functions and architecture boundaries

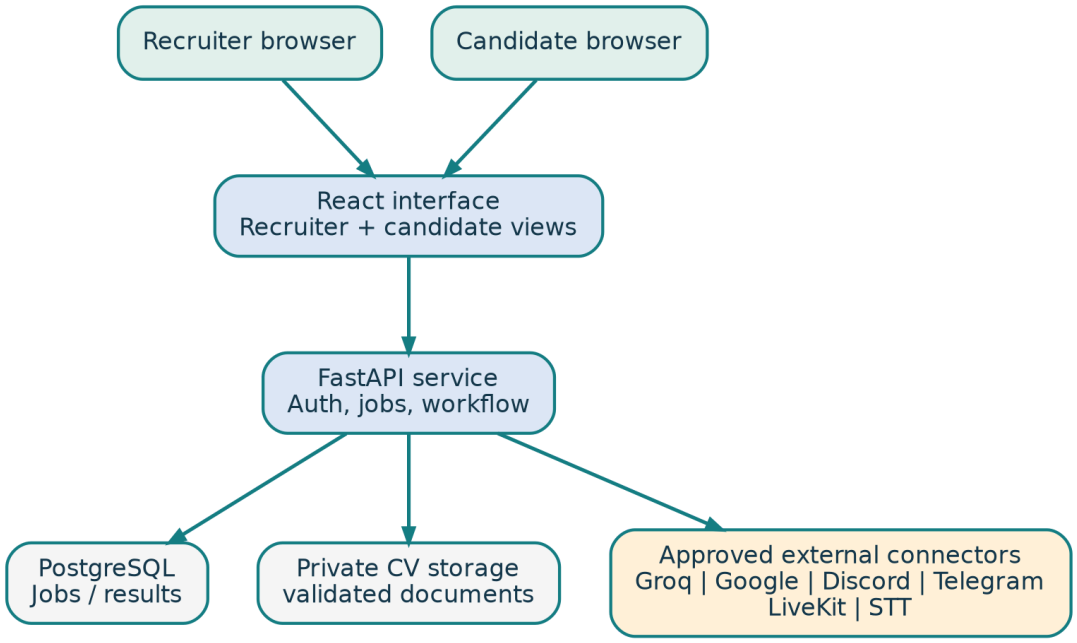


Figure 2. Logical system context and external service boundaries.

Layer	Purpose / boundary
Frontend	React/Vite UI; recruiter workflow, candidate invitation entry, media widgets, screening views and reports.
Backend API	Python/FastAPI with Pydantic validation, authentication, job ownership, persistence and provider boundaries.
Agentic layer	LangGraph state, steps, conditional routes, retries and human approval points for applicable AI tasks.

Layer	Purpose / boundary
Storage	PostgreSQL for jobs, criteria, CV records, scores, invitations, transcript/assessment records; private file storage for documents.
AI providers	Configured Groq API for JD/CV; alternate local provider selection may exist, but is not required for the current intended demo.
Interview / STT	LiveKit for two-person audio/video; AssemblyAI and local Whisper experimental paths must be distinguished by actual runtime readiness.
Other integrations	Discord MCP publishing, manual LinkedIn copy, optional Telegram, and separate Google Forms/Drive intake adapter.

2.3 Assumptions and dependencies

- Reviewers use a supported modern desktop browser with internet access for external providers and a working Docker engine for packaged deployment.
- Recruiter has acknowledged any externally processed job/CV/interview data before provider transmission.
- Required provider API keys are configured locally and never committed; free-tier quota is not guaranteed and provider failure must be handled.
- Job and candidate records, approved criteria revisions, interview invitations and transcripts use explicit ownership and association checks.
- Discord MCP and Telegram Bot API publishing each require explicit private destination approval and provide real transport receipts. Google Forms/Drive one-shot import requires scoped OAuth, one-response owner authorization and a trusted ledger. Manual LinkedIn copy is not automated posting.
- Optional biometric and monitoring extensions require explicit consent, synthetic evaluation identities, and a separate privacy/risk review; these extensions do not replace core BID functions.

2.4 Constraints and human oversight

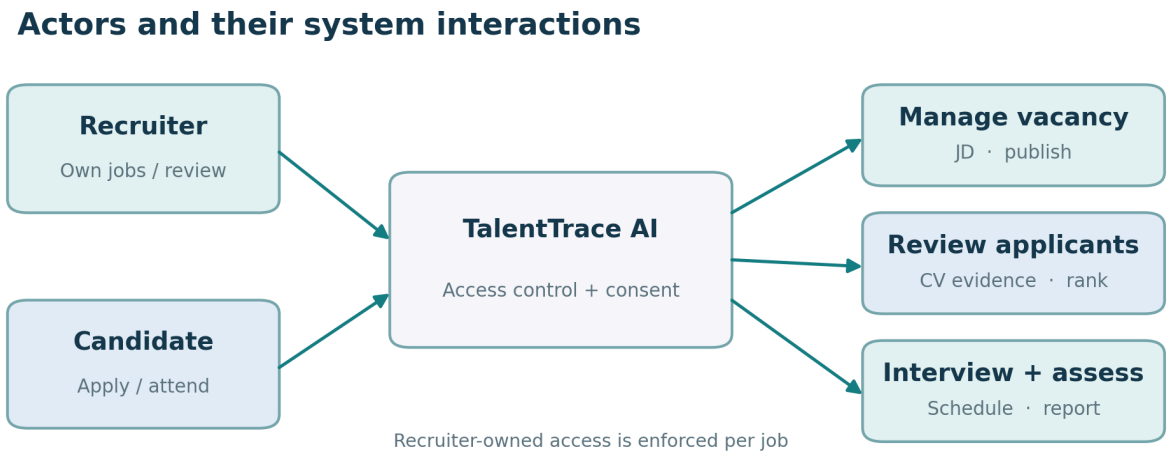


Figure 3. Actor roles and functional interactions.

The system must not automatically reject or hire candidates based on LLM output. Recruiters explicitly review key criteria, candidate selection, external publishing destinations, interview

scheduling and sensitive data transfers. Failures or incomplete evidence must be surfaced rather than silently treated as successful analysis.

3. Functional requirements

Each row describes expected behavior and a proposed acceptance outcome. “Source” differentiates official BID mandatory items, chosen innovation and trainer-reported additions. Test IDs are stable cross-references for the future UAT matrix.

3.1 Access, role separation and job ownership

Priority: M + T | Source: BID N7–N9; reported trainer guidance

Recruiter and candidate roles must have separate capabilities; all job-scoped records must be protected.

ID	System requirement (shall)	Acceptance / verification	Test ID
AUTH-01	Authenticate recruiter and allow secure logout.	Authenticated recruiter sees owned workspace; logout invalidates the session.	TC-01
AUTH-02	Restrict recruiter job, CV and assessment reads/writes to the owning recruiter.	Another recruiter cannot access or change records by guessing identifiers.	TC-02
AUTH-03	Issue one-use, interview-scoped candidate invitation and session.	Used, expired or revoked links cannot be re-used to create another session.	TC-13
AUTH-04	Exclude candidate role from scores, reports, recruiter-only prompts and other job records.	Candidate API and UI have no recruiter-only data.	TC-17

3.2 JD creation, revision and criteria approval

Priority: T | Source: Reported verbal trainer guidance; project design

The recruiter creates and refines a grounded JD conversationally while retaining controlled job identity.

ID	System requirement (shall)	Acceptance / verification	Test ID
JD-01	Provide a conversational JD drafting form from role, experience, skills and policy.	A real configured model generates a draft that recruiter can review.	TC-03
JD-02	Preserve Job Reference across JD refinements; increment revision after successful save.	A targeted skill change advances revision but not Job Reference.	TC-04
JD-03	Display and persist generated/revised JD and conversation history.	Refresh restores text and correct revision.	TC-04
JD-04	Require human confirmation of screening criteria after creation or meaningful revision.	Screening is blocked against unapproved criteria.	TC-05

ID	System requirement (shall)	Acceptance / verification	Test ID
JD-05	Default to a configured JD model, disclose external cloud processing before its first use and keep the historical model pinned to each conversation. CV-processing permission is independent.	Groq is visible in the compact composer; one-time JD disclosure precedes cloud transfer; CV consent is not reused for JD.	TC-03

3.3 Job publishing review

Priority: T / extension | Source: Reported verbal trainer guidance; original BID excludes posting

Publishing remains human-reviewed and should distinguish true automatic delivery from manual copying.

ID	System requirement (shall)	Acceptance / verification	Test ID
PUB-01	Recognize recruiter publishing commands and show the saved JD preview.	“publish” opens review for current saved job without regenerating text.	TC-06
PUB-02	Require explicit destination selection and confirmation.	All destinations begin unchecked; no message is sent merely by preview.	TC-06
PUB-03	Publish approved job content to configured Discord via the existing MCP adapter with explicit confirmation, provider receipt and bounded duplicate-attempt safeguards.	Private-channel post contains original Job Reference; the actual Discord message ID/channel are verified independently.	TC-07
PUB-04	Publish to a privately configured Telegram channel using the Bot API; keep LinkedIn and other destinations manual copy-only.	Verified Telegram message ID and actual text; LinkedIn shows “Copied — not posted”; no destination is published by preview alone.	TC-06
PUB-05	Handle independent publication destinations, explicit reviewer choice, masked credentials and partial-failure or uncertain delivery correctly.	One provider failure never reports another provider as posted; no blind re-post or secret logging.	TC-06, TC-07, TC-22

3.4 CV intake, document extraction and OCR

Priority: M + T | Source: BID F1; trainer-reported Google intake

The system accepts job-associated batches, isolates records and extracts readable document content.

ID	System requirement (shall)	Acceptance / verification	Test ID
CV-01	Upload multiple valid PDF CVs and supported DOCX documents with type/size validation.	Correct job receives durable batch identity; invalid files fail safely.	TC-08
CV-02	Extract text from digital PDFs and supported document formats.	Text and page references are available for downstream analysis.	TC-09
CV-03	Use configured OCR for scanned or mixed-content PDF pages.	A synthetic scanned page yields readable text with source-page provenance.	TC-09
CV-04	Preserve recruiter/job ownership and deduplicate native/imported CV identities.	Switching jobs never displays another job's applications.	TC-10
CV-05	Import specifically authorized Google Forms/Drive submissions through a bounded, trusted, idempotent one-shot operator workflow, not by merely configuring the public form URL.	Server-side timestamp filter returns one consented target response; only its CV is read; stored ledger and Home count one imported application.	TC-11
CV-06	Reject corrupt/empty/unreadable files without crashing the rest of the batch.	Per-file failure is visible and remaining documents continue safely.	TC-08
CV-07	Validate a Google response's stable identity, candidate consent, owned Job Reference, single file ID, format, size, signature and cryptographic integrity before import.	Wrong job, historical response, ambiguous page, invalid consent/format or repeat import cannot leak or double-count data.	TC-11, TC-23

3.4.1 Application notifications and analytics

Priority: T | Source: Reported verbal trainer guidance

Application activity should be countable and explicitly acknowledged without leaking candidate content.

ID	System requirement (shall)	Acceptance / verification	Test ID
NOT-01	Display scoped application totals: received, new, viewed and available per selected job.	Correct counts reflect distinct application identities rather than scoring attempts.	TC-21
NOT-02	Allow the recruiter to explicitly acknowledge displayed new applications.	Acknowledged items persist as viewed after refresh; later arrivals remain new.	TC-21
NOT-03	Prevent cross-recruiter/job leakage and count duplicate intake records once.	Switching jobs changes totals; another recruiter sees none of these records.	TC-21

3.5 F1 — CV screening and candidate ranking

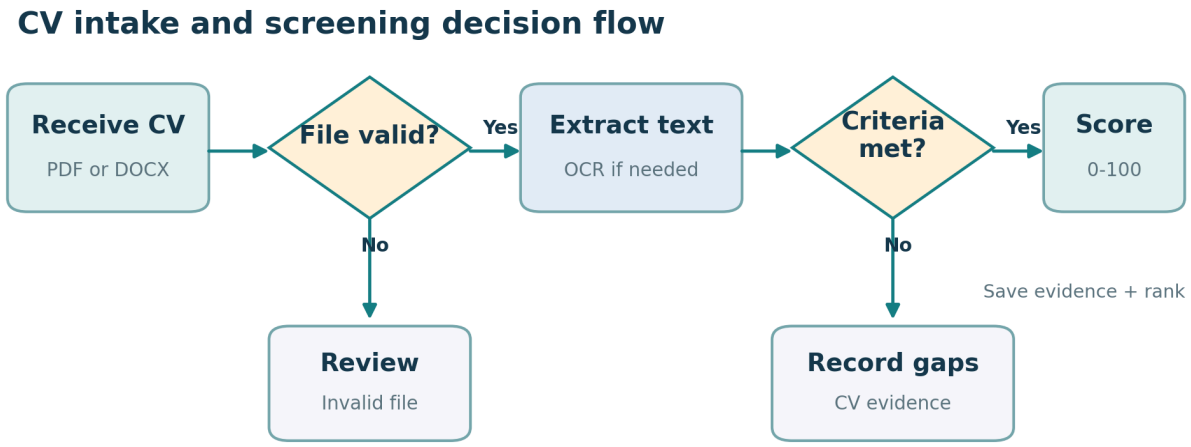


Figure 4. F1/F2 CV processing and screening decisions.

Priority: M | Source: BID §5.2.1 F1

Automated, explainable requirement-based screening is mandatory.

ID	System requirement (shall)	Acceptance / verification	Test ID
F1-01	Accept job requirements: required/preferred skills, qualifications, experience and selection criteria.	Approved job profile can be saved and loaded.	TC-05
F1-02	Extract structured contact, education, work, skill, certification and project attributes from a CV.	Source-backed profile displays where information came from.	TC-09
F1-03	Use semantic matching rather than keyword equality alone.	Meaning-equivalent synthetic skill claims receive appropriate match state.	TC-10
F1-04	Classify each requirement as met, partially met or unmet, with evidence.	Recruiter can inspect reasons and source excerpts.	TC-10
F1-05	Rank all scored applicants for the same job and criteria revision.	Order aligns with defined score and eligibility rules.	TC-10
F1-06	Fail safely on unusual formats, missing sections and scanned files.	Batch outcome distinguishes successful, partial and failed records.	TC-08

3.6 F2 — AI-based candidate scoring

Priority: M | Source: BID §5.2.1 F2

Each candidate needs explainable, weighted suitability scores and durable provenance.

ID	System requirement (shall)	Acceptance / verification	Test ID
F2-01	Score technical skills, relevant experience and qualifications separately.	Results contain all three dimensions or explicit missing-evidence states.	TC-10
F2-02	Allow configurable scoring weights per approved job profile.	Changing approved weights changes computed aggregate deterministically.	TC-10
F2-03	Compute an aggregate and display the component breakdown.	UI shows total, component values and applied weight basis.	TC-10
F2-04	Provide rationale and CV-backed evidence for each component.	Evidence references correspond to actual CV content.	TC-10
F2-05	Persist scores, provider/model provenance, consent and criteria revision.	Reload retrieves same results and audit context.	TC-12
F2-06	Handle provider errors and reproducibility limits explicitly.	No false score appears after failed inference; repeat variance is documented.	TC-12

3.7 Scheduling and invitation workflow

Priority: T | Source: Reported trainer guidance; BID scheduling excluded

This extension requires explicit recruiter confirmation and independently controlled candidate access.

ID	System requirement (shall)	Acceptance / verification	Test ID
SCH-01	Offer only eligible candidates from an approved current criteria revision for scheduling.	Unapproved or stale candidates cannot be booked silently.	TC-13
SCH-02	Let recruiter select time zone, start and duration and preview final start/end time.	Preview displays the exact job, candidate and time window.	TC-13
SCH-03	Require recruiter acknowledgement and explicit final confirmation before booking.	One click at confirmation creates one durable booking.	TC-13
SCH-04	Keep booking and invitation separate; do not claim email was sent unless configured and accepted.	Reload persists booking; delivery clearly reports not configured/accepted.	TC-13
SCH-05	Enforce interview join window and candidate invitation expiry/revocation.	Late, early, reused or revoked access is rejected.	TC-14
SCH-06	Offer scoped candidate self-scheduling from approved invitation and available slots, without bypassing recruiter ownership or booked-slot checks.	Synthetic candidate books allowed slot; recruiter sees durable booking after refresh; no claim of real-email proof from fake SMTP.	TC-13

3.8 F3 — integrated video interview

Priority: M | Source: BID §5.3.1 F3

Interview sessions must be real two-person video/audio sessions, associated to the selected job and candidate.

ID	System requirement (shall)	Acceptance / verification	Test ID
F3-01	Create or open a scheduled interview associated with job and candidate identity.	Recruiter selects saved eligible candidate and correct interview.	TC-13
F3-02	Allow recruiter and candidate to join from separate browsers securely.	Both independently reach Connected; no cross-interview leakage.	TC-14
F3-03	Publish and receive live audio/video media from each party.	Both see/hear permitted participant media.	TC-14
F3-04	Support mute, camera toggle and explicit end/leave controls.	State changes appear correctly and tracks stop when leaving.	TC-14
F3-05	Route interview audio to the authorized transcription pipeline.	Consented, attributed audio reaches F4; no processing without consent.	TC-15
F3-06	Handle media interruption and disconnection without corrupting saved session data.	User sees recoverable error, retained transcript remains intact.	TC-14
F3-07	Persist participant/session start and end lifecycle.	Authorized reviewer retrieves durable session metadata after reload.	TC-14

3.9 F4 — speech recognition and transcript

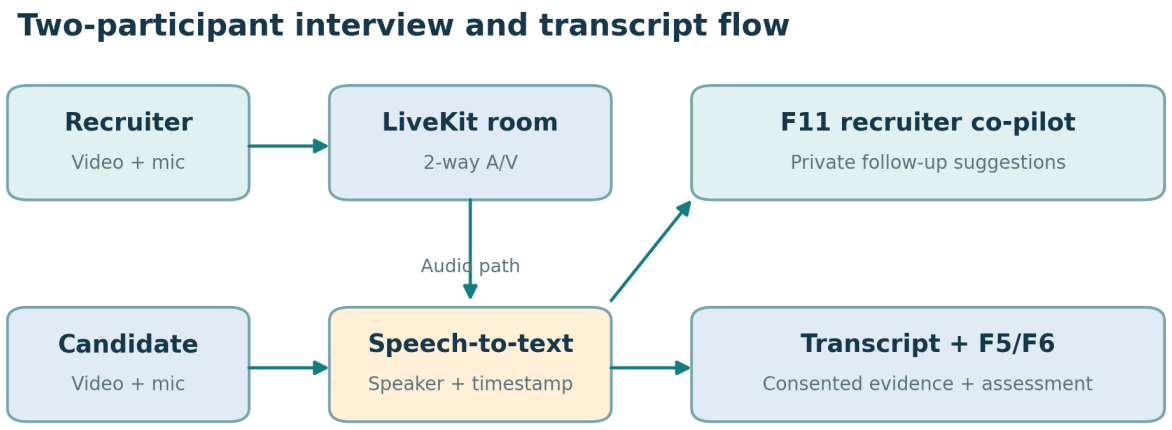


Figure 5. F3/F4/F11 live interview, transcript and co-pilot data flow.

Priority: M | Source: BID §5.3.1 F4

F4 must produce real speaker-attributed, timestamped, retrievable transcript turns.

ID	System requirement (shall)	Acceptance / verification	Test ID
F4-01	Capture consented speech from both interview participants during the session.	Each speaker is represented in the transcription capture path.	TC-15
F4-02	Identify recruiter vs candidate by channel/role and timestamp each utterance.	Transcript entries include speaker and start/end or turn time.	TC-15
F4-03	Display live transcription with limited perceptible delay.	Live caption display updates while synthetic speech is played.	TC-15
F4-04	Persist structured transcript turns and retrieve them after interview ends.	Reload returns complete recorded turns.	TC-15
F4-05	Handle silence, overlapping speech, accents and STT provider failure gracefully.	UI displays partial/uncertain result, never fictional text.	TC-15
F4-06	Disclose external STT transfer and respect explicit caption/storage consent or withdrawal.	Unconsented data is not transmitted; withdrawal stops further collection.	TC-17

3.10 F5 — interview Q&A analysis

Priority: M | Source: BID §5.4.1 F5

Analyze actual transcript content for job-relevant insight without inventing unobserved evidence.

ID	System requirement (shall)	Acceptance / verification	Test ID
F5-01	Segment interviewer questions and candidate responses into linked Q&A units.	Retrieved transcript yields correct links and source turns.	TC-16
F5-02	Evaluate relevance, technical accuracy, completeness and articulation.	Each evaluated answer includes component ratings or an explicit not-assessable state.	TC-16
F5-03	Map each question/answer to approved job requirements.	Coverage map distinguishes covered and unaddressed competencies.	TC-16
F5-04	Reference transcript turns in evaluation justifications.	Reviewer can locate cited utterances.	TC-16
F5-05	Handle follow-ups, multipart answers and non-assessable small talk.	Segmentation avoids fabricating questions or answers.	TC-16

3.11 F6 — consolidated candidate assessment

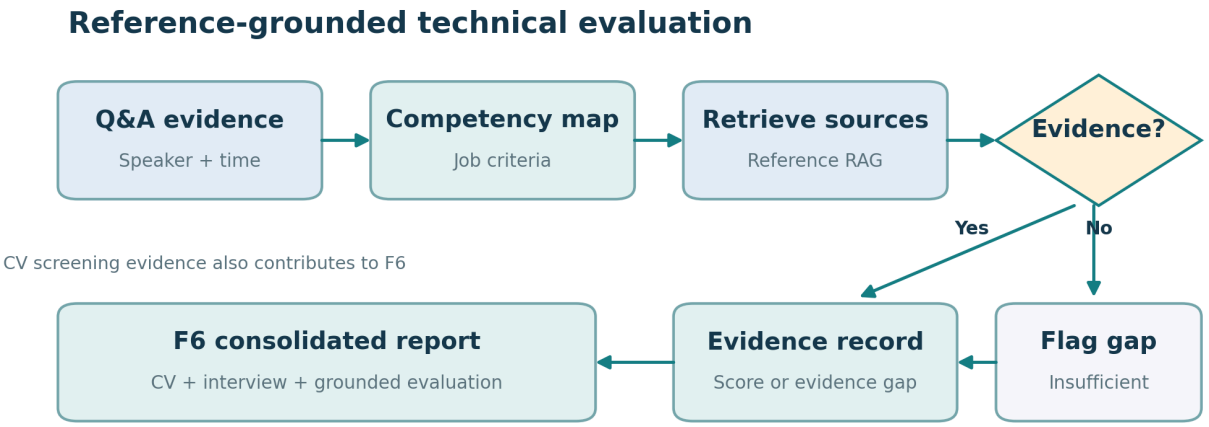


Figure 6. F5/F6 reference-grounded evaluation flowchart.

Priority: M | Source: BID §5.4.1 F6

Generate a unified reviewer-friendly report with traceable conclusions and honest evidence gaps.

ID	System requirement (shall)	Acceptance / verification	Test ID
F6-01	Combine F1/F2 screening evidence with F5 analysis for the same candidate, job and criteria revision.	No mixing across jobs, candidates or revision versions.	TC-18
F6-02	Present overall suitability and explain its component evidence, not just one opaque score.	Report shows strengths, weaknesses, gaps and score origins.	TC-18
F6-03	Keep missing transcript/interview evidence explicitly unavailable.	CV-only report never pretends a completed interview occurred.	TC-18
F6-04	Support readable print or export format.	Reviewer exports/prints an assessment with no secret leakage.	TC-18
F6-05	Compare multiple candidates for the same job using comparable dimensions.	Side-by-side view preserves job/revision association.	TC-18
F6-06	Link conclusions to CV or transcript source material.	Reviewer can trace each important finding.	TC-18

3.12 F11 — Dynamic Interviewer Co-Pilot

Priority: I | Source: BID §6 innovation; approved project concept documented in WBS

Optional recruiter-only prompts target missing coverage or shallow answers during a consented interview.

ID	System requirement (shall)	Acceptance / verification	Test ID
F11-01	Observe authorized F4 turns and approved job requirements without candidate exposure.	Candidate API/DOM never receives co-pilot data.	TC-17
F11-02	Detect unaddressed job competencies and insufficient interview depth.	Suggestion links to current coverage state.	TC-17
F11-03	Display timely follow-up suggestions in recruiter interface only.	Recruiter sees prompt; candidate sees none.	TC-17
F11-04	Require separate assistant consent; permit withdrawal and dismiss.	Withdrawal halts further suggestions; stale polls do not resurrect dismissed prompt.	TC-17
F11-05	Keep the innovation rationale and integration architecture documented.	Rationale identifies beneficiary, trade-offs and distinctness from F1–F10.	TC-20

3.13 Other reported and stretch capabilities

Priority: O / T | Source: BID F7–F10; verbal trainer guidance

Additional capabilities must not be presented as core success substitutes.

ID	System requirement (shall)	Acceptance / verification	Test ID
EXT-01	Allow separate optional screen sharing with visible on/off status and participant grant.	Sharing starts only on user action and stops on Leave.	TC-14
EXT-02	Show consented candidate page visibility/fullscreen activity as a coarse indicator, not cheating proof.	Recruiter receives last-reported status only; withdrawal stops reporting.	TC-17
EXT-03	If F7 is enabled, provide consent-gated synthetic face enrollment/comparison within defined identity and retention controls.	Comparison uses authorized synthetic subjects; real-person identity assertions require separate safeguards.	TC-19
EXT-04	Allow separate configuration of approved JD/CV AI providers with no unauthorized fallback.	Selection, provenance and consent are independently enforced.	TC-03
EXT-05	Provide separate optional controls for F8 CV-claim flags, F9 interview scoring and F10 interview-quality assessments when selected.	Every displayed optional score has a source and explicit eligibility rule; suppressed options do not influence F1–F6.	TC-18

4. User interfaces and external interfaces

4.1 UI inventory



Figure 7. Conceptual recruiter interface wireframes (not application screenshots).

UI ID	Screen / actor	Key actions	Associated requirements
UI-01	Sign in / recruiter	Authenticate, navigate Home, logout.	AUTH-01, AUTH-02
UI-02	Recruiter Home	Select job, review activity, use next-action navigation.	JD, CV, SCH, N25
UI-03	JD Assistant	Resumable narrower chat workspace, larger scrollable message body, compact JD model selector, revision, criteria review and publish preview.	JD-01–05, PUB-01
UI-04	Candidate Screening	Native and trusted imported batch review, OCR/profile extraction, screening and evidence with job-scoped criteria.	CV, F1, F2
UI-05	Scheduling	Eligible candidate, review time/duration, confirm, inspect booking.	SCH-01–05
UI-06	Recruiter interview	Invite, join, media controls, consent, captions, co-pilot.	F3, F4, F11
UI-07	Candidate interview	Accept invitation, consent, join/leave; no recruiter-only fields.	AUTH-03/04, F3
UI-08	Assessment / reports	View CV+interview evidence, compare/export.	F5, F6

4.2 Software and communications interfaces

Interface	Contract / constraint	Privacy and failure mode
Groq API	Configured LLM provider for JD creation and CV analysis;	Clear one-time JD cloud disclosure; separate CV

Interface	Contract / constraint	Privacy and failure mode
	availability may depend on free quota.	external-processing authorization, no silent provider fallback; quota/failure surfaced.
AssemblyAI STT	Separate, permissioned microphone streams produce speaker-associated timestamped captions and final turns.	Independent transcription and storage consent, provider timeout policy and restricted transcript retrieval.
LiveKit Cloud	Two-participant WebRTC media connection with scoped grants and room ownership.	Secure transport; fresh invitations; no leak of tokens/room credentials; avoid exceeding free plan.
Google Forms/Drive	Published candidate application form, Drive CV upload and read-only OAuth; single approved response maps to the exact owned job.	Time- and identity-scoped API read; validate consent, file identity, hash and idempotent import. Scheduled sync is a separate contract.
Discord MCP	Explicitly reviewed private destination with MCP stdio publication and provider message receipt.	Fixed configured webhook, masked credentials and delivery readback; no implicit repeat on uncertainty.
LinkedIn / Telegram	Telegram Bot API destination receives reviewed JD; LinkedIn is a manual copy/share interface.	Review chosen channel, verify message identity and do not equate copied content with posted content.
Tesseract OCR	Native OCR and English language data for scanned PDFs.	Bounded file/page processing and source provenance.
PostgreSQL	Durable job, candidate, score, schedule, session and assessment storage.	Restrict access by recruiter and job; migrations preserve existing data.

4.3 Hardware interfaces

No special hardware is mandatory for the synthetic demonstration. Browsers may use camera/microphone devices when authorized, but generated media is preferable in automated tests. The application relies on ordinary network, CPU, RAM and storage facilities of the host and its Docker runtime.

5. Non-functional requirements

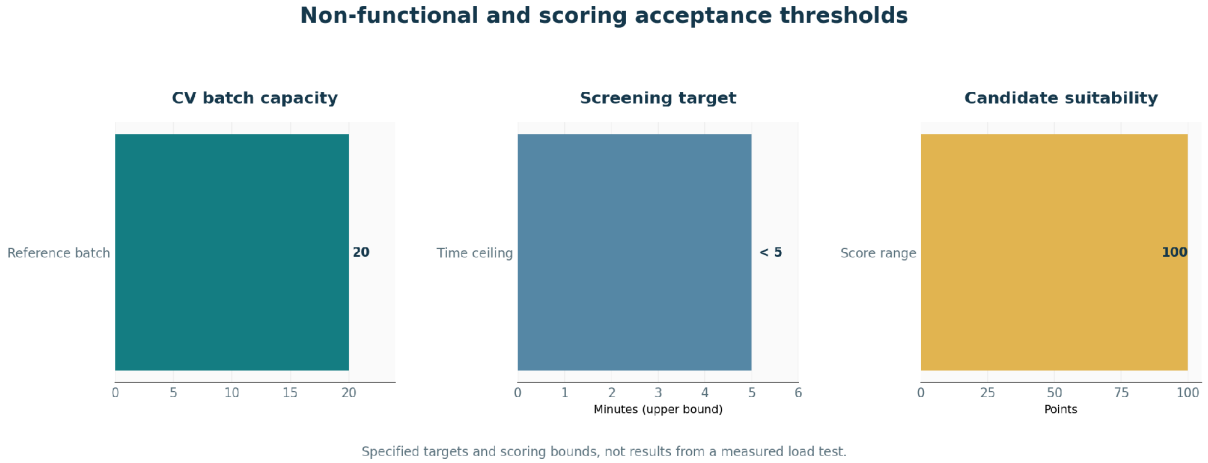


Figure 8. Acceptance targets and score scale (specified values, not benchmark results).

IDs N1–N26 and their intent are taken from BID §7; numerical thresholds are not invented where the BID only specifies qualitative targets. Each is proposed for objective acceptance evidence where feasible.

BID ID	Quality area	Verifiable requirement
N1	Performance	20-CV batch parsing/screening/scoring/ranking measured and documented; report elapsed time, environment and failures.
N2	Performance	Live transcription has minimal noticeable lag; measure representative synthetic turnaround.
N3	Performance	Video/audio stays usable during a typical interview; disconnect tests recorded.
N4	Performance	Non-AI endpoints respond promptly; long AI work exposes progress/status instead of blocking UX.
N5	Performance	At least one interview coexists with background CV processing without failure.
N6	Reliability	AI/provider failure produces understandable recoverable error without data loss.
N7	Security	API keys and database credentials only in environment; sanitized .env.example; never committed.
N8	Security	Validate all inputs and file type/size; reject malformed payloads.
N9	Security	Candidate cannot access interviewer scores, ranking, assessment or co-pilot.
N10	Security	Use secure transport for networked APIs/media; document localhost development exceptions.
N11	Maintainability	Pin dependencies; avoid unsupported/unnecessary packages.
N12	Security	Public errors never disclose stack, paths, prompts, tokens or secrets.
N13	Privacy	Use provided approved sample CVs or synthetic data; no unconsented real CV.
N14	Privacy	Exclude CVs, recordings, personal screenshots, transcripts and database dumps from Git.
N15	Privacy	Anonymize screenshots, reports, presentations and logs.
N16	Privacy	Minimize collected data; avoid long-term raw AV without justified need.

BID ID	Quality area	Verifiable requirement
N17	Privacy	If biometric F7 attempted, explicit consent, retention, security and low-confidence disclosures.
N18	Privacy	Document external recipients and candidate data sent to each.
N19	Privacy	Provide guarded candidate and linked-data deletion workflow.
N20	Fairness	Use job-relevant criteria only; no protected-characteristic inputs; document bias limits.
N21	Delivery	Verify documented Docker startup on a clean machine with no cached artifacts.
N22	Operations	Structured privacy-safe logs for significant events and failures.
N23	Maintainability	Modular well-named code and clear responsibilities.
N24	Documentation	README and API/OpenAPI cover architecture, setup, configuration, usage and limits.
N25	Usability	Workflow navigable without oral coaching; understandable progress and error messages.
N26	Persistence	Jobs, CV records, scores, transcripts and reports survive app restart.

6. Design constraints and technical platform

6.1 Mandated BID technology

Component	BID constraint	TalentTrace AI proposed / current architecture
Backend	Python and FastAPI + Pydantic	Python 3.12 / FastAPI with validated API schemas.
AI orchestration	Meaningful LangGraph graphs with state, nodes, routes	Use explicit graph steps where reasoning is multi-stage; document real graphs, not one-node wrappers.
Containerization	Full Docker Compose containerization	React frontend, FastAPI backend, PostgreSQL and supporting tools defined in Compose.
Source control	GitHub and meaningful incremental commits	Private repository; one active evaluation branch for continuing work.
Frontend	Trainee choice, justified	React, TypeScript and Vite; explain trade-offs in README.

6.2 Packaging and deployment requirements

- D1/D3: Define frontend, backend, database and required support services in Docker Compose, including dependencies and networks.
- D2/D8: Provide one documented startup command plus explicit build, stop and reset commands; distinguish prepared-runtime restart from clean-machine installation.

- D4: Use environment variables, ship a sanitized .env.example and never commit populated secrets.
- D5: Use persistent volumes for data that must survive container restart.
- D6/D7: Verify a clean-machine build and maintain a suitable .dockerignore and bounded image sizes.
- G1–G8: Keep Git repository hygiene, README completeness, innovation rationale, evolving WBS and release tag.

6.3 Explicit architectural and privacy constraints

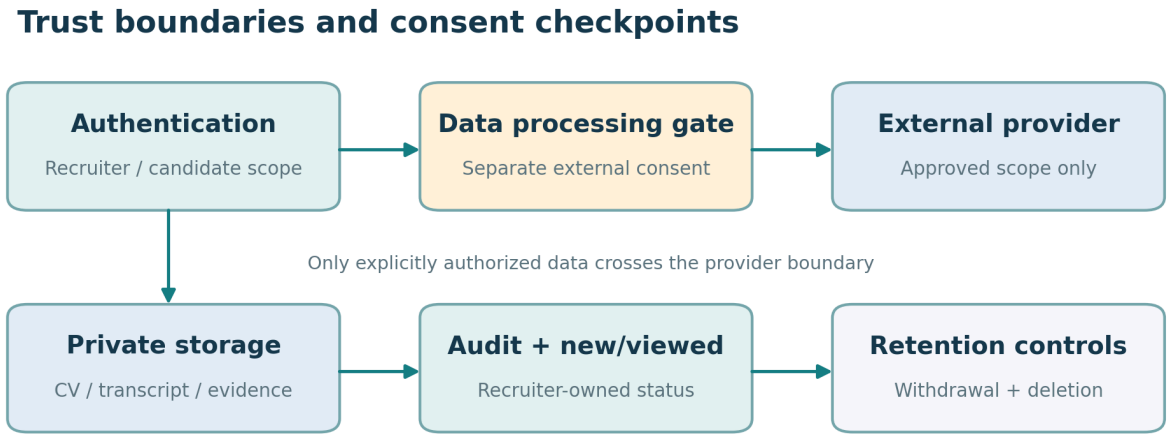


Figure 9. Authorization and privacy boundaries across services.

Provider selection should not overwrite a pinned historical conversation. CV and JD consent states must be independent; changing job/revision requires a new consent decision when external processing changes context. Recruitment recommendations always remain subject to human review. Trial media and biometric prototypes are not evidence of production readiness.

7. Verification and acceptance plan

Acceptance evidence shall be collected from application interactions, authenticated API/database outputs, approved external-provider receipts, real media/transcript runs and persistence checks. Synthetic candidate data shall be used. A unit test is a supporting control and is not interchangeable with end-to-end evidence.

Test ID	Scenario	Expected observation
TC-01	Recruiter sign-in / logout	Valid synthetic recruiter signs in; logout invalidates access.
TC-02	Ownership boundary	Two recruiters/jobs; cross-owner API and UI access denied.
TC-03	Groq JD creation	Real synthetic JD inference, provider/consent clear; save and reload.
TC-04	JD revision	Add preferred Docker skill; same reference, higher revision; reload.
TC-05	Criteria confirmation	Human review required; stale revision rejected for screening.
TC-06	Publishing preview	Publishing preview, explicit destination choice; manual

Test ID	Scenario	Expected observation
		LinkedIn preview/copy clearly not published.
TC-07	Discord delivery boundary	Real approved Discord MCP post to private channel; verify actual message ID, channel and original Job Reference.
TC-08	Native batch upload	Valid & invalid PDFs/DOCX; per-file feedback; correct job.
TC-09	Digital + scanned OCR	Mixed PDF extraction; actual OCR page evidence recorded.
TC-10	F1/F2 semantic scoring	Approved job, synthetic CVs, real Groq, score breakdown, evidence, ranking.
TC-11	Google imported batch	One server-side filtered Google response; exactly one signed/hashed CV; job-scoped trusted import and no read of nine historical submissions.
TC-12	Provider provenance/reload	Same saved score/model, consent reference, criteria revision after refresh.
TC-13	Recruiter booking	Reviewed recruiter booking plus candidate self-selection of invited slot; persistence after refresh; distinguish fake SMTP from actual email.
TC-14	F3 two-browser media	One recruiter + one candidate genuinely connect, publish/receive and Leave.
TC-15	F4 live STT	Generated speech over actual interview media -> timestamped transcript, restart retrieval.
TC-16	F5 transcript analysis	Real transcript -> Q&A pairs, 4 dimensions, requirement gap and references.
TC-17	F11 and privacy	Recruiter-only suggestions, consent, dismiss, withdrawal and candidate API/DOM exclusion.
TC-18	F6 report	CV+interview report, no invented missing evidence, export/compare.
TC-19	Face prototype boundary	Synthetic-only offline check, low-confidence and Windows vault-disabled state.
TC-20	Delivery/UAT	Clean-machine Docker Compose start, logs, README, Git tag and full manual walkthrough.
TC-21	Notification acknowledgement	New, received, viewed totals remain job-scoped and persist after acknowledgement.
TC-22	Telegram actual delivery	One private Bot API message, verified message ID and Job Reference; no duplicate publication.
TC-23	Imported application notification	Approved one-shot import updates Home to 1 received / 1 new / 1 available / 1 trusted

Test ID	Scenario	Expected observation
		import; unrelated job remains unchanged.

8. Prioritization and release plan

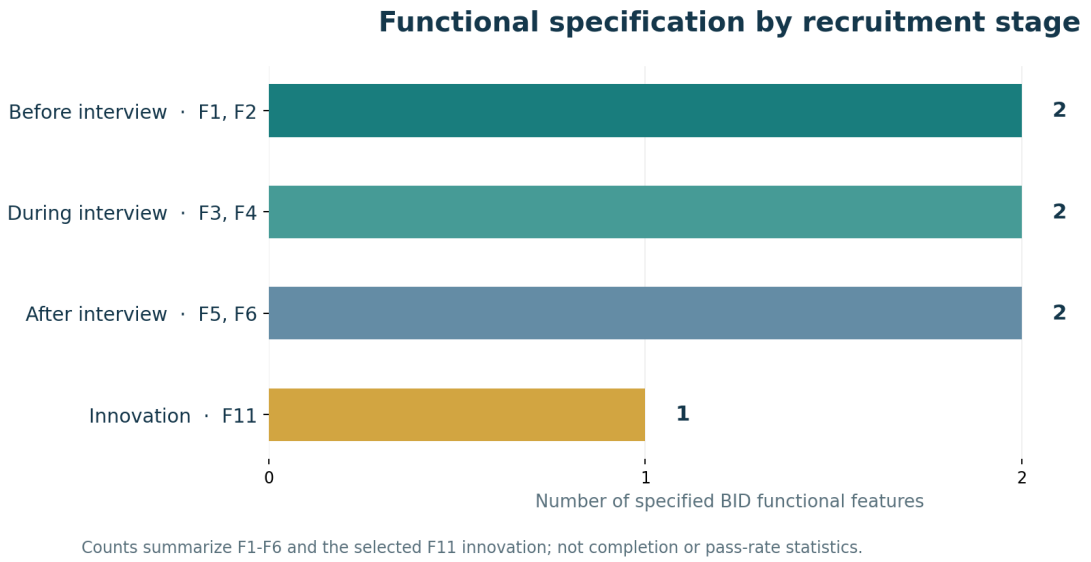


Figure 10. BID functional-feature allocation by lifecycle stage.

The BID organizes the compulsory workflow into six core functions (F1–F6), one selected innovation (F11), and associated quality and delivery obligations. Acceptance proceeds through the phase gates below while preserving the same recruiter-owned job and synthetic candidate evidence.

Gate	Required demonstrated behavior	Dependency
R1 — Baseline	Docker launch, login, private database and saved jobs.	AUTH, N7–N9, N21/N26
R2 — Pre-interview	Same-job JD revision and real approved external publication; synthetic Form submission + single-response Drive import; native/linked CV screening and evidence.	R1; provider and consent
R3 — Interview	Scheduling/invitation, two-party LiveKit F3, F4 live caption persistence.	R2; LiveKit and STT readiness
R4 — Assessment	F5 Q&A and F6 cross-source report; F11 recruiter co-pilot.	R3; durable transcript
R5 — Final acceptance	Document complete SRS; preserve reproducible evidence and record remaining N1, N19, N21 and connected interview acceptance limitations without inflating PASS.	R1–R4

9. Requirements traceability and acceptance mapping

9.1 Source traceability

SRS group	Authoritative or supplemental origin	Priority
F1/F2	BID §5.2.1; WBS module 3	M
F3/F4	BID §5.3.1; WBS modules 4–5	M
F5/F6	BID §5.4.1; WBS module 6	M
F11	BID §6 innovation + selected WBS module 7	I
AUTH, N1–N26, Docker	BID §§7–8	M
JD, publishing, Google intake, notifications, scheduling, provider choice	Reported verbal trainer guidance and project decisions	T
Face scan, optional activity, screen sharing	Reported verbal trainer guidance and/or BID F7–F10 stretch	T/O

9.2 Acceptance evidence register and verification method

The matrix below defines the acceptance artifact associated with each feature. A requirement is evaluated through its named use case, output evidence and protected data boundary; test execution records are managed outside the SRS.

Requirement group	Acceptance evidence type	Expected observer outcome
JD creation + refinement	Saved conversation, job revision, criteria snapshot	Stable Job Reference; modified preferred skill without modifying required criteria.
Controlled publishing	Provider receipt, approved destination and post readback	Revised job content delivered only after explicit destination review; manual copy distinguished.
External candidate intake	One filtered Forms response + Drive file + import record	Authorized job-matched synthetic application enters private CV storage exactly once.
Applications dashboard	Authenticated job-scoped application list and counters	Received/new/viewed counts agree with unique stored or imported CV identities.
F1 candidate screening	Parsed file extract, OCR evidence, ranked eligible candidates	Required criteria and CV source evidence shown for every candidate outcome.
F2 candidate scoring	Saved dimension weights, scores and explanations	Reproducible score linked to criteria version and cited CV evidence.
Scheduling	Recruiter booking and candidate invitation record	Candidate sees only permitted slots; booking persists on recruiter view.
F3 media session	Two-party LiveKit browser session	Recruiter and candidate exchange genuine real-time media within the correct room.
F4 transcription	Finalized labeled turns + retrievable transcript	Speaker and time attribution survive refresh under active storage consent.
F5 interview analysis	Persisted Q&A evaluation + references	Question/answer assessments trace to transcript turns and technical reference sources.

Requirement group	Acceptance evidence type	Expected observer outcome
F6 consolidated report	Job-specific printable comparison and report	CV and interview scores, evidence and decision-support summary remain associated.
F11 recruiter co-pilot	Recruiter-only suggestion feed and privacy checks	Consent-gated prompts do not expose private tips in the candidate interface.
Privacy and deployment	Access-control tests, Compose commands, persistent records	Cross-owner denial, consent boundaries and persistent data behavior are demonstrated.

10. Risk controls and assumptions

Risk ID	Design or delivery risk	Control and acceptance condition
RK-01	Interpretation of verbally reported extensions	Retain BID priority; obtain trainer confirmation where extension conflicts with scope.
RK-02	Multi-provider latency or temporary outage	Fail safely with explicit retry policy, bounded timeout and preserved job identity.
RK-03	CV personal data processed by cloud provider	Use explicit scope-specific consent, restricted transfer and synthetic demo data.
RK-04	External publication to unintended destination	Require reviewed destination, fixed webhook/channel and receipt validation.
RK-05	Duplicate Forms response or repeated import	Bind stable source ID to job; make write idempotent and verify file hash.
RK-06	Ambiguous Forms response selection	Apply a server-side time/identity filter, reject ambiguity and never scan unrelated submissions.
RK-07	Candidate identity or cross-job disclosure	Use recruiter-owned job authorization, signed or scoped candidate invitation, and deny-by-default APIs.
RK-08	Unreliable interview network or media devices	Keep room lifecycle explicit; provide visible error and bounded reconnect strategy.
RK-09	STT error or unknown speaker attribution	Preserve timestamps, label uncertain content and separate captions from finalized stored turns.
RK-10	LLM hallucination or unsupported conclusion	Ground outputs in CV/transcript/reference evidence and show uncertainty.
RK-11	Docker or token portability between machines	Use documented Compose profiles, secret isolation, repeatable initialization and persistent volumes.

Appendix A. Trainer-requested additions

This register records fourteen developer-reported trainer additions for requirement traceability. In case of conflict, the written BJIT Business Information Document remains the primary authority.

Item	Reported guidance
T01	Conversational JD refinement with human review and job reference.
T02	A visible stable Job Reference for generated JD.
T03	Chat-driven publishing review with multiple destinations.
T04	Strict per-job CV/application isolation.
T05	New/received/viewed notification counts and analytics.
T06	Recruiter-reviewed interview scheduling with start/end time before email.
T07	Candidate-specific secure access to scheduled interview.
T08	Recruiter-only interview follow-up suggestions (F11 innovation).
T09	Isolation for simultaneous interviews and interview-dependent processing.
T10	Face scan online/physical: exploratory prototype; verification policy TBD.
T11	Local vs API provider selection, with explicit consent for external processing.
T12	Optional candidate tab/fullscreen activity indicators, not cheating proof.
T13	Participant-controlled screen sharing.
T14	Modern, simple Home and clear recruitment navigation.

Appendix B. References and authority

Ref	Description
[R1]	BJIT, Business Information Document: AI-Powered Recruitment & Interview Intelligence Platform, version 2, 34 pages. Primary source for mandatory core, quality and delivery constraints.
[R2]	BJIT, SRS Template.doc, 12-page supplied example. Used for section ordering, revision table, requirement ID / acceptance / test columns and UI inventory.
[R3]	Amazing Lunch Indicator Software Requirements Specification, 58-page supplied sample. Used as a pattern for introduction, stakeholders, constraints, traceability and release prioritization; its restaurant requirements are not imported.
[R4]	TalentTrace AI WBS planning material (available v1.0.1–v1.0.3 task-level workbook excerpts). Used for phase and work-package mapping; not evidence of implementation.
[R5]	Private GitHub repository shafi-mahmud/TalentTrace-AI and developer-reported office acceptance, screenshots, and local commit outcomes through 28 October 2026 (including 483c251 UI/publishing and 8a00a6c one-shot Google import). Live local Git state not directly inspected by this document editor.
[R6]	Verbal trainer recommendations as recollected by the developer in project discussions; not formal signed clarifications.

Appendix C. Additional visual models

The additional diagrams below adapt the sample SRS approach to TalentTrace AI. They specify interactions and acceptance sequencing without implying that a visual schematic is itself a test result.

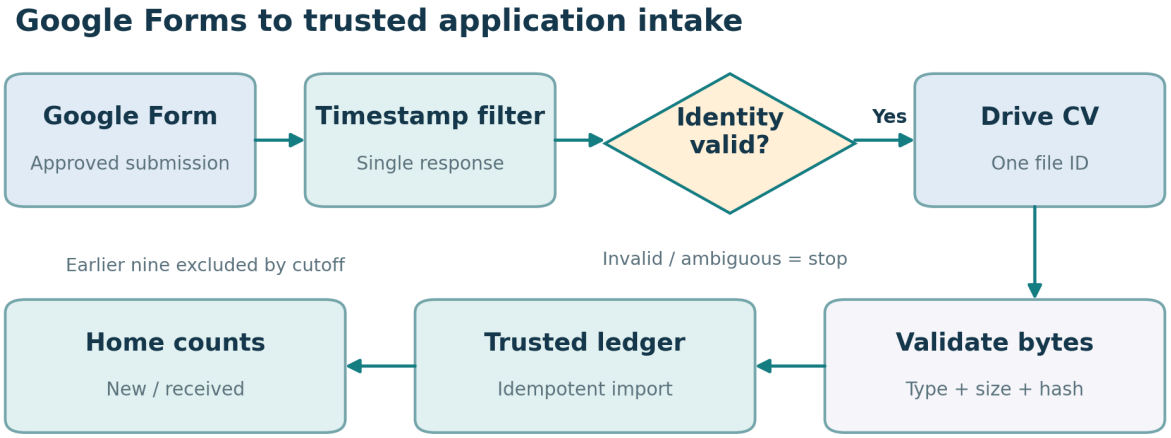


Figure 11. One-response Google Forms / Drive trusted-intake flowchart.

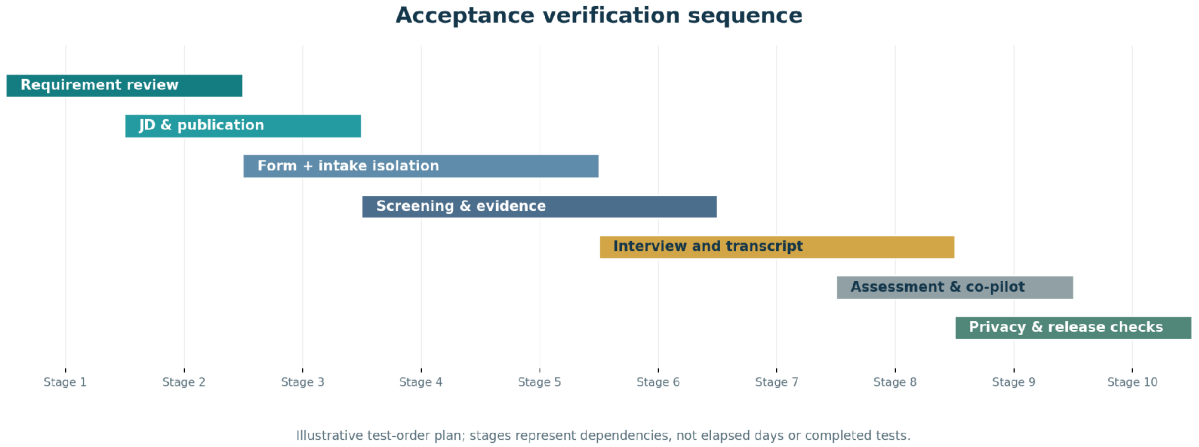


Figure 12. Acceptance test sequencing diagram (illustrative order, not a completion timeline).